



PROJECT CONTEXT FOR CURSOR

Project Name: Ashgate Real Estate Platform

Primary Developer: Tyrese Muigai

1. OVERVIEW

We are building a **multi-module real estate web application** for **Ashgate Ltd**, designed to digitize property listings, property management, land development advisory, and sustainability (carbon credits).

The system serves both **public users** (browsing listings) and **authenticated users** (landlords, tenants, admins).

It must be **secure, scalable, visually modern**, and optimized for both desktop and mobile access.

2. USERS & ACCESS ROLES

Role	Description	Access Scope
Admin	Ashgate management staff (Tyrese & uncle). Approve landlord accounts, manage listings, oversee the system.	Full access across all modules.
Landlord / Agent	Own or manage properties. Use admin interface to manage buildings, tenants, rent collection, and maintenance.	Access limited to their properties and tenants.
Tenant	Occupies property units. Uses web app to pay rent, view bills, and make maintenance requests.	Access tenant dashboard only (frontend).
General User	Public visitor. Browses property listings, land advisory info, and sustainability features. No login required.	Full access to public content only.

✓ Landlords and admins both log in through the **admin interface** (Django Unfold UI).

✓ Tenants and general users interact through the **React frontend**.

3. MODULES & FUNCTIONALITY

Module	Purpose	Key Features
Property Listings	Showcase all properties to public users.	Search/filter, map view, image/video gallery, property details, SEO-ready pages.
Property Management	For landlords and tenants.	Building/unit registration, rent collection (Mpesa API), utilities billing, maintenance requests, tenant approvals.
Land Development Advisory	For landowners seeking feasibility insights.	Submit land details, receive ROI and development suggestions, investor connection requests.
Carbon Credits Integration	Sustainability feature tied to properties or developments.	Emission calculator, offset purchase logging, "Ashgate Green Certified" badges.
Admin & Analytics	Oversight and management.	Approvals, dashboards, reports, user management, role controls, content management.

4. TECHNOLOGY STACK

Frontend (Public + Tenant):

- React + Next.js 15
- Tailwind CSS
- React Hook Form + Yup (validation)
- Redux Toolkit or Zustand (state)
- Recharts / Chart.js (dashboards)
- Google Maps API
- Mpesa payment popup integration
- SEO metadata for listings

Backend:

- Django 5 + Django REST Framework

- JWT Authentication
- Role-based permission system (Admin, Landlord, Tenant)
- Django Unfold for a modern admin panel
- REST API endpoints for frontend consumption
- Integration with Mpesa Daraja API, Cloudinary/AWS S3

Database & Infrastructure:

- PostgreSQL (main relational database)
- AWS S3 / Cloudinary (media storage)
- Redis (optional, for caching and Celery tasks)
- Hosting: Vercel (frontend) + Render/AWS (backend)
- Cloudflare (SSL + CDN)

Security & Compliance:

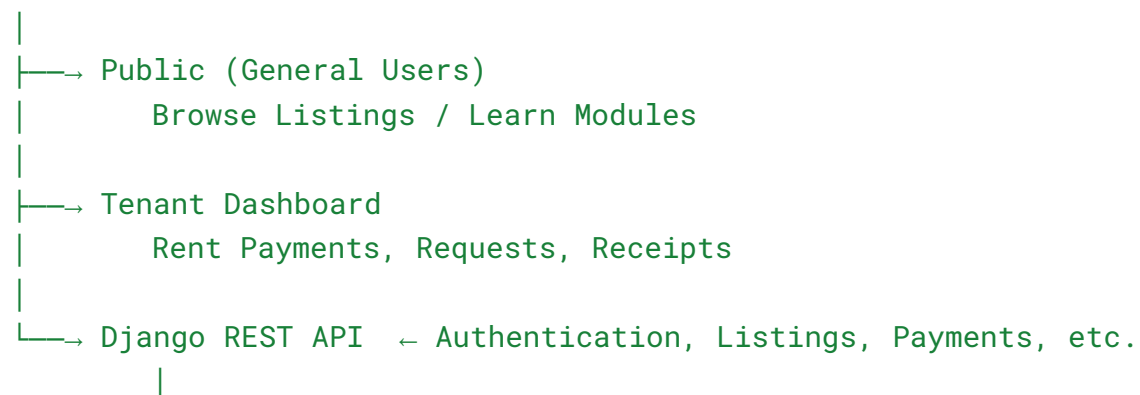
- HTTPS enforced
- Password hashing (Django default)
- Data protection compliant with Kenyan Data Protection Act
- Encrypted environment variables

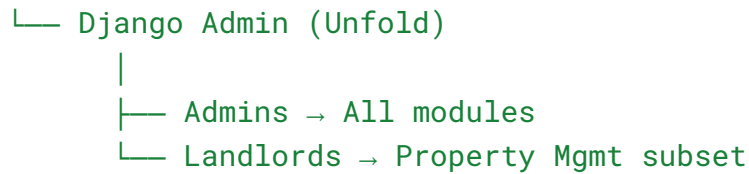
5. ADMIN INTERFACE DESIGN (Django Unfold)

- Responsive sidebar navigation, branding with Ashgate logo/colors.
- Admin roles can access all data.
- Landlords log in here but only see property management data filtered to their account.
- Integrated charts for rent summaries, occupancy, and maintenance stats.

6. SYSTEM ARCHITECTURE FLOW

Frontend (React + Next.js)





7. DEVELOPMENT STAGES (Milestone-Based)

Stage 1 — Property Listings + Property Management (Ksh 20,000)

- Setup Django + DRF + PostgreSQL
- JWT authentication & role models
- React frontend for public listings
- Property CRUD, search, filters, map integration
- Property management module (landlords, tenants)
- Mpesa sandbox integration for rent
- Unfold admin theme installation & configuration

Stage 2 — Land Advisory + Carbon Credits (Ksh 20,000)

- Land development advisory module: submission forms, feasibility uploads, investor connections
- Carbon credit module: emission estimator, offset tracking, certification badges
- Admin tools for managing developers, carbon providers, reports
- Public pages explaining sustainability features

Stage 3 — Admin Interface + Finalization (Ksh 30,000)

- Finalize Unfold dashboard branding, permissions, charts
- Full analytics/reporting pages
- SEO setup, static content, contact forms
- Deployment (Vercel + Render/AWS)
- SSL, backups, user documentation
- Final testing and delivery

8. EXPECTED OUTCOME

By completion, the system will provide:

- A public, SEO-optimized real estate platform for listings and sustainability.
- Secure, modern admin dashboard for Ashgate and landlords.
- Tenant self-service tools for payments and maintenance.
- Land advisory and carbon offset integration that differentiates Ashgate in the market.

9. DEVELOPMENT GUIDELINES

- Follow clean architecture and modular folder structure (**apps/** for Django modules).
- All API endpoints documented in Swagger/OpenAPI.
- Use environment variables for secrets (Mpesa, DB, etc.).
- Reusable React components for both tenant and public views.
- Commit regularly with clear messages and feature branches per stage.

That's the full brief. Cursor should interpret this as the **system context** and maintain this scope and stack throughout development.